Data warehouse: Specification of business processes in SigmaPol Rental network

Specification of business processes

1. Business goals of the organization

“SigmaPol Rental” is a car rental network operating in the five cities in Poland: Warsaw, Kraków, Bydgoszcz, Gdańsk and Poznań. Each city has several rental points located in its districts. Cars are rented only in rental points (the company does not rent online). The main goal of the company is to achieve the highest possible profit from renting cars. To achieve this goal Michał Schumacher (CEO) wants to **increase the car occupation time**, i.e. to make sure that cars do not stay unrented: he assumes that the fleet utilization should **not be lower than 80%**, so that a car stays idle for no more than 20% of the time. He monitors two main metrics: the utilization rate of cars in individual rental points (the number of days cars were rented divided by the number of days they were available) and the number of idle days (days when an available car was not rented) within a month. He assumes that the company is successful if **the utilization rate in every rental point is not lower than 80%** and the number of idle days falls from month to month. Currently his biggest problem is to answer the question why in the analyzed month the number of idle cars has increased or decreased.

The most important business issues are renting cars and servicing the fleet. Each car has a make, model, colour, year of production, fuel type (petrol, diesel, LPG, hybrid or electric), mileage and type (suv, van, combi, sedan, coupe or sport). Cars are rented by private customers, and each rental is handled by an employee of a rental point. Each rental point is managed by one person.

Michał Schumacher would like to analyze the influence of holidays, weekends and the summer season on the utilization of cars. Moreover he wants to know which car makes, models, fuel types and car types are rented most often. In addition, the CEO wonders if the time cars stay unrented depends on the employees who handle customers, and if the age and the mileage of cars, the time since their last service and the services (including those for unexpected problems) influence the time they stay unrented.

2. Business processes

Car rental

a. A general description of the business process and a description of the performance metrics generated by this process, possible current analytical problems.

The customer enters the rental point and chooses a car and the rental period. The employee, logged in to the rental system, checks the availability of the car, verifies the customer’s identity document and driving licence. A new customer fills in a form with personal data, which the employee then enters into the client EXCEL sheet (an existing client is identified by PESEL number). The employee assigns the chosen car (identified by its licence plate) to the rental and records the mileage of the car. The system generates a rental agreement with the date of issue, the planned return date and the cost, calculated from the daily price of the car and the number of days. The customer pays the cost. When the car is returned, the employee records the actual return date and the mileage of the car; if the car is returned later than planned, the cost is recalculated for the actual number of days. Metrics: car utilization rate, number of idle days, number of rentals, number of rental days, average rental length and rental revenue. Analytical problems: explaining month-to-month changes of utilization and finding the periods, rental points, makes and fuel types for which cars stay idle.

b. Typical questions

- Compare the number of rentals in terms of car fuel types.

- What are the most frequently rented car makes and models in this month?
- What is the utilization rate of cars in individual rental points compared to the 80% target?
- Which cars stayed idle (unrented) for the longest time in this month?
- Compare individual employees in terms of the number of rentals they handled.
- Compare the utilization rate of cars in the last month with the average annual utilization rate.
- Compare the utilization rate of cars during holiday periods and weekends with other days.
- What are the numbers of idle days of cars in individual rental points?
- What is the average length of rentals in individual cities?
- How many rentals were returned later than planned in individual rental points?

c. Data

All rental data are extracted from the rental system - “RentMaster”. The system stores information about the employee who handled the rental (thanks to logging in), the rental point, the car (licence plate, make, model, colour, year of production, fuel type, type, current mileage, daily price), the PESEL number of the client, the date of issue, the planned and actual return dates, the mileage of the car before and after the rental and the cost of the rental. In addition, data about clients (name, surname, PESEL, age and gender derived from PESEL, driving licence number and phone number) are stored in an EXCEL sheet: the client fills in a form and then the employee enters the data into the sheet. Everything else is taken from the “RentMaster” system.

Fleet servicing

a. A general description of the business process and a description of the performance metrics generated by this process, possible current analytical problems.

A car is directed to a routine (periodic) service after every 15,000 km or 12 months from its last service, whichever comes first. A car can also be directed to a service when an unexpected problem occurs, e.g. after an accident (type accident) or a breakdown or when the brakes have to be changed (type malfunction); such a problem is usually reported by the employee when the car is returned. An employee of the rental point sends the car to a workshop; during the service the car is unavailable for rental. After the service the employee registers it in the system as a new record of the service history: the type of the service (routine, accident or malfunction), the optional description, the mileage of the car, the dates of the arrival in and return from the service and the cost of the service. The last service of a car results from this history. Metrics: number of services, number of services for unexpected problems, days in service (unavailability), cost of services, mileage since the last service, time since the last service and number of cars exceeding the service interval. Analytical problem: the unknown influence of routine servicing, services for unexpected problems and the mileage since the last service on the time cars stay unrented (idle time).

b. Typical questions

• Which cars exceeded 15,000 km since the last service?
• Which cars have not been serviced for more than 12 months?
• Compare the number of cars serviced in the last month in individual rental points.
• What is the average mileage since the last service for each fuel type?
• Compare the number of routine services and services for unexpected problems (accident, malfunction) in individual rental points.

• Which car makes and models need services for unexpected problems (accidents, malfunctions such as brake changes) most often?

c. Data

Data about services are also fetched from “RentMaster” (table Service), which keeps for every service its type (routine, accident or malfunction), the optional description, the dates of the arrival in and return from the service, the mileage of the car and the cost of the service. The last service of a car results from this history.